

MAE 271A Project

Calibration of an Accelerometer Using GPS Measurements

Ethan Lien
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Abstract

The goal of this project is to implement a minimum variance estimator, specifically a Kalman filter algorithm, to accurately estimate the system state and accelerometer bias provided noisy accelerometer reading and GPS measurements. The performance of the designed algorithm proved to be successful, minimizing the error between the true state and estimated state for a single realization of initial state and bias. Moreover, through Monte Carlo simulation, theoretical properties of the Kalman filter are satisfied, satisfying expected value checks, variance checks, and orthogonality and independence checks.

1 Introduction

Accelerometers are common sensors that provide readings of acceleration for a given system. From these readings, other important states of a system can be determined, such as position and velocity. However, When incorporating these devices into a control scheme, using the reading directly will prove to be a very naive approach. As with all instruments, accelerometers outputs are subject to noise and, more importantly for us, may prove to have a steady state bias that affects all measurements it produces. Fortunately, readings from devices like these can be supplemented by other devices, like a GPS, to provide more accurate measurements.

Our goal in this project is to simulate a noisy, biased 1D accelerometer and use synthetic GPS measurements to create a minimal variance estimator, specifically the Kalman filter algorithm, to accurately determine the true state of our system, as well as the bias inherent to our accelerometer. For consistency between trials, the same model for acceleration will be used throughout 30 second intervals:

$$a(t) = 10 \sin(\omega t), \omega = 0.2 \text{ rad/sec} \quad (1)$$

After a functioning program is produced, Monte Carlo simulations will be performed to verify theoretical properties of the Kalman filter.

2 Theory

The notion of the Kalman filter algorithm is to create a minimum variance estimator that reports an estimate of a system's state provided measurements and a dynamic model of the system of interest. Provided a discrete-time Gauss-Markov system

$$\begin{aligned} x_{k+1} &= \Phi_k x_k + \Gamma_k w_k, \quad x_0 \sim N(\bar{x}_0, M_0) \\ z_k &= H_k x_k + v_k \end{aligned}$$

where

x is our state

z is our measurement

w_k and v_k are Gaussian white-noise processes with known covariances
 w_k , v_k , and x_0 are independent of one another

we can produce a set of equations that result in an estimation of the system's state. Specifically, there are two main portions of the algorithm. Firstly

$$\begin{aligned}\bar{x}_{k+1} &= \Phi_k \hat{x}_k \\ M_{k+1} &= \Phi_k P_k \Phi_k^\top + \Gamma_k W_k \Gamma_k^\top\end{aligned}\tag{2}$$

is our propagation step, which uses our system dynamics to predict how our state and error variance change between our measurements. Next we have

$$\begin{aligned}K_k &= M_k H_k^\top (H_k M_k H_k^\top + V_k)^{-1} \\ \hat{x}_k &= \bar{x}_k + K_k (z_k - H_k \bar{x}_k) \\ P_k &= (I - K_k H_k) M_k (I - K_k H_k)^\top + K_k V_k K_k^\top\end{aligned}\tag{3}$$

as our update step. This set of equations uses a measurement and the predicted state from above to produce our conditional mean $\hat{x}_k$ and error variance P_k , which represent a more accurate estimation of our system's state.

This algorithm has many beneficial consequences due to its construction. Because x, v , and z are Gaussian in distribution, our conditional mean $\hat{x}$ is both our maximum likelihood estimate and our minimum variance estimate of our state. Moreover, because estimate is a function of our measurement and our state is mean-symmetric and unimodal, $\hat{x} = \mu$ (the mean of the state) will minimize a quadratic loss function and produce minimal variance. That is to say, the error variance generated by the Kalman filter will be the smallest possible variance for our system.

3 Implementation

The implementation of a Kalman filter requires that we have knowledge of both the dynamics and measurement processes of our system of interest. Two state properties that we are taking interest in are position and velocity. Because our accelerometer provides us an acceleration over time, we can use an Euler integration process to discretize our dynamics.

$$\begin{aligned}v(t_{j+1}) &= v(t_j) + a(t_j) \Delta t \\ p(t_{j+1}) &= p(t_j) + v(t_j) \Delta t + a(t_j) \frac{\Delta t^2}{2}\end{aligned}\tag{4}$$

where Δt is the sampling period of our sensor, 1/200Hz.

While we now have a model for how the the state due to an acceleration, let us now derive a model for the acceleration as reported by an accelerometer. Our model needs to take into account: a bias, $b \sim N(0, 0.01(\text{meters/sec}^2)^2)$ and Gaussian white noise $w(t) \sim N(0, 0.0004(\text{meters/sec}^2)^2)$.

$$a_c(t_j) = a(t_j) + b + w(t_j)\tag{5}$$

We can discretize Equation 5 and subtract the true state equations provided by Equation 4 to obtain our dynamics equation for our Kalman filter:

$$\begin{bmatrix} \delta p(t_{j+1}) \\ \delta v(t_{j+1}) \\ b(t_{j+1}) \end{bmatrix} = \begin{bmatrix} 1 & \Delta t & -\frac{\Delta t^2}{2} \\ 0 & 1 & -\Delta t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \delta p(t_j) \\ \delta v(t_j) \\ b(t_j) \end{bmatrix} - \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \\ 0 \end{bmatrix} w(t_j) \quad (6)$$

where

$$\begin{aligned} \delta p(t_0) &= p(t_0) - p_c(t_0) \sim N(0, M_0^p) \\ \delta v(t_0) &= v(t_0) - v_c(t_0) \sim N(0, M_0^v) \end{aligned}$$

Now that we have the dynamics for the differences, our algorithm will no longer rely on the specific trajectory produced by the true acceleration experienced by the system.

Next we produce the measurement equation for the Kalman filter. We have GPS measurements defined by

$$\begin{aligned} z^p(t_i) &= p(t_i) + \eta^p(t_i) \\ z^v(t_i) &= v(t_i) + \eta^v(t_i) \end{aligned} \quad (7)$$

where measurements occur at a frequency of 5Hz, and η^p and η^v are zero-mean with known variance. In a similar process to the state dynamics, we subtract the true position and velocity from Equation 7 as

$$\begin{bmatrix} \delta z^p(t_i) \\ \delta z^v(t_i) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \delta p(t_i) \\ \delta v(t_i) \\ b(t_i) \end{bmatrix} + \begin{bmatrix} \eta^p(t_i) & 0 \\ 0 & \eta^v(t_i) \end{bmatrix} \quad (8)$$

Now that we have our dynamics and measurement equations in reference to a difference is state, the estimate that our Kalman filter produces will be as difference as well, defined by

$$\hat{\delta x}(t_i) = \begin{bmatrix} \hat{\delta p}(t_i) \\ \hat{\delta v}(t_i) \\ \hat{b}(t_i) \end{bmatrix} = \begin{bmatrix} \hat{p}(t_i) - p_c(t_i) \\ \hat{v}(t_i) - v_c(t_i) \\ \hat{b}(t_i) \end{bmatrix} \quad (9)$$

4 Results

For an individual realization of the designed Kalman filter algorithm, the following state estimation was produced over a 30 second period:

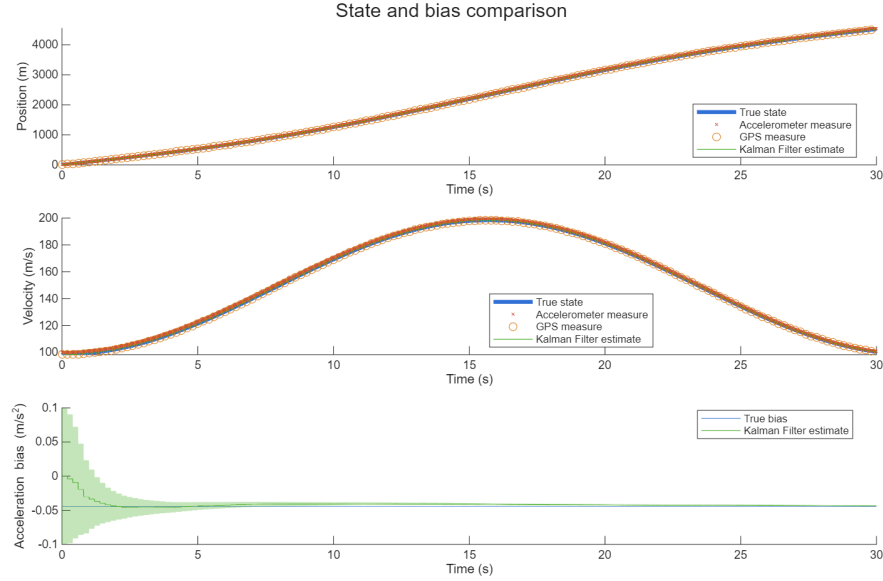


Figure 1: State position, velocity, accelerometer bias for a single iteration of the Kalman filter algorithm. True, accelerometer, measured, and estimated state are presented with one sigma bound generated from algorithm.

From a glance, Figure 1 demonstrates that our estimated state and bias converges to the true state, regardless of the error between the truth model and the accelerometer reading. To better observe the convergence, the estimate error is isolated and plotted.

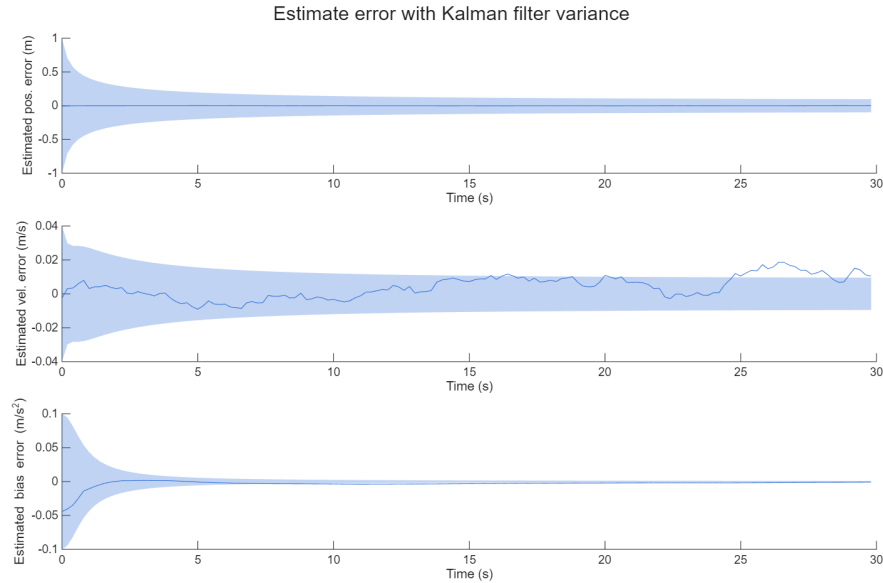


Figure 2: Difference between updated state estimate and true state with one sigma bound.

The Kalman proves to be sufficient for this realization of the accelerometer model. To verify that the algorithm works for any number of realizations, we will check the performance of an ensemble of realizations and check them against theoretical properties of the Kalman filter.

The first property is that the average error between the update estimate and the truth model is 0. This is because our minimum variance estimator produces conditional

mean $\hat{x} = \mu$, so our expected value for our error should be the mean value of our state variables' distribution. For a realization l , we will define an ensemble average $e^l(t_i)$ for N_{ave} realizations to be

$$e^{ave}(t_i) = \frac{1}{N_{ave}} \sum_{l=1}^{N_{ave}} e^l(t_i) \approx 0, t_i \in [0, 30] \quad (10)$$

Computing the above for $N_{ave} = 2500$, we can plot the resulting average over time.

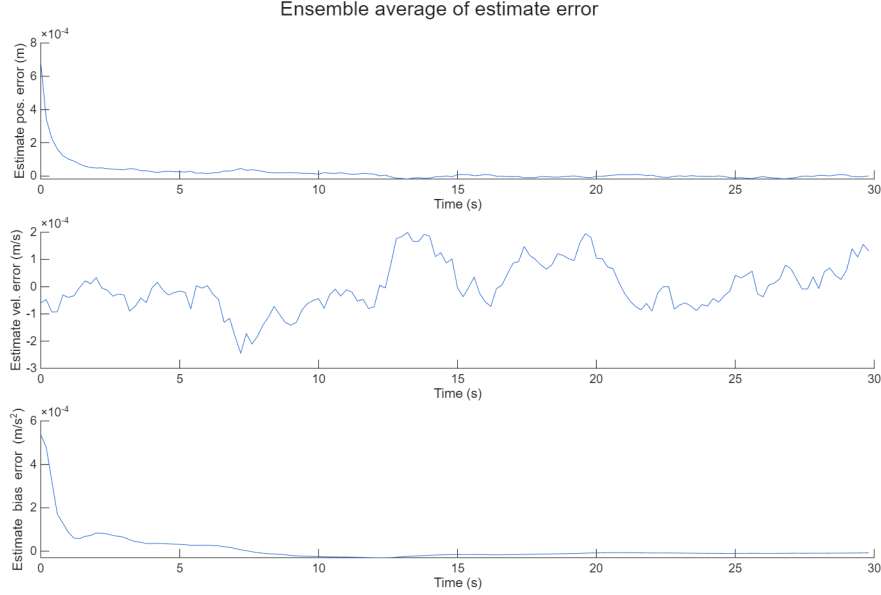


Figure 3: Ensemble average of estimate error for $N_{ave} = 2500$ realizations of accelerometer model.

It appears that the Kalman filter we designed satisfies this check, as the average error is very small over the entire 30 second interval. Next, we will compare our Kalman filter error variance $P(t_i)$ to an empirically calculated P^{ave} using our average estimate error:

$$P^{ave}(t_i) = \frac{1}{N_{ave} - 1} \sum_{l=1}^{N_{ave}} [e^l(t_i) - e^{ave}(t_i)][e^l(t_i) - e^{ave}(t_i)]^\top \quad (11)$$

The computed $P^{ave}(t_i)$ is expected to converge to that of the Kalman filter. Computing $P(t_i) - P^{ave}(t_i)$, we expect the difference to be zero.

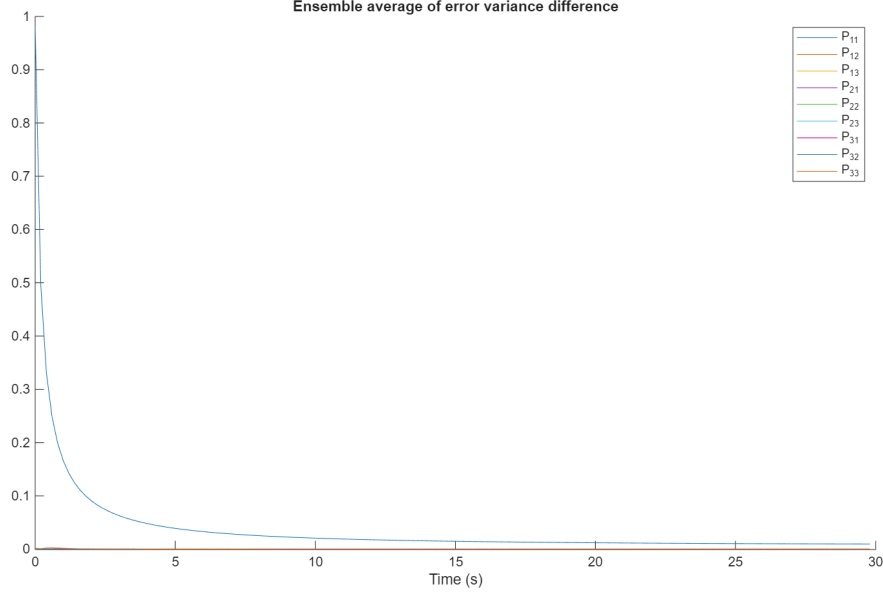


Figure 4: Time plot of each element of $P(t_i) - P^{ave}(t_i)$.

From Figure 4, the difference converge to values close to zero. There is an exception, with the (1,1) term starting with a relatively large difference before decaying towards zero. This could be attributed to the $P(t_i)$ computed by the Kalman filter begin independent of the measurement of the state. The estimated error variance will not consider what is being measured, the actual estimate error is usually quite small, thus producing very small covariances on average. Thus, this term is different at the start but eventually decays as the estimated variance is propagated through time.

We can evaluate another property for the Kalman filter. It is theorized that the expected value of the estimate error and the estimate transpose is zero. We will verify this property by seeing if

$$\frac{1}{N_{ave}} \sum_{l=1}^{N_{ave}} [e^l(t_i) - e^{ave}(t_i)] \hat{x}(t_i)^\top \approx 0 \quad (12)$$

hold true for our realizations.

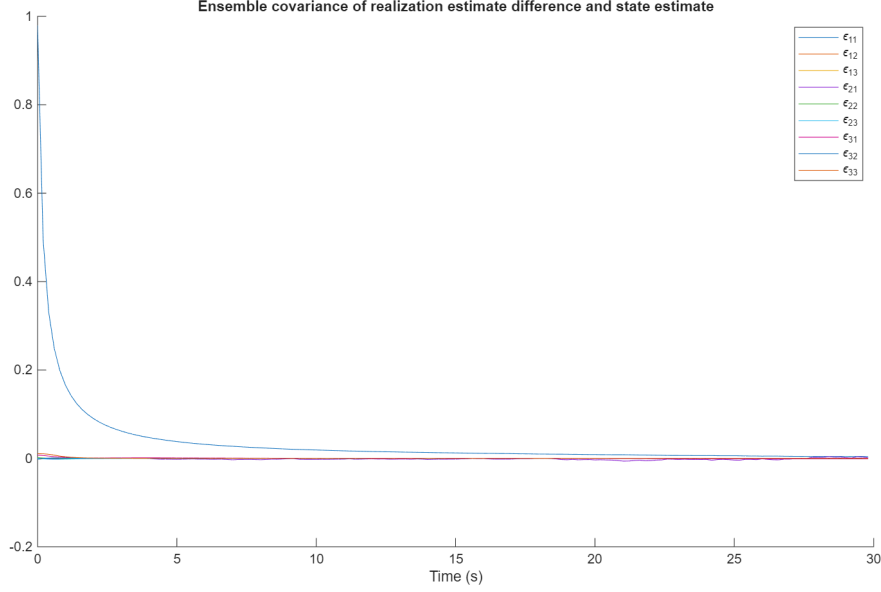


Figure 5: Check of the orthogonality of estimate error with the estimate in reference to average estimate error.

We see similar results to those of Figure 4, where most terms of the resulting matrix are zero, demonstrating orthogonality. Again, there is exception in the (1,1) term of the matrix. Though it starts from a vastly different value than the other terms, its convergence to zero over time shows that this orthogonality principle still hold for our designed filter.

A final check for our algorithm would be to check the independence of our residuals from the update step of the Kalman filter. Because all of our probability functions are Gaussian white noise, the residuals should maintain orthogonality with itself at different time instances. Thus, we will determine if

$$\frac{1}{N_{ave}} \sum_{l=1}^{N_{ave}} r^l(t_i) r^l(t_m)^\top \approx 0 \quad \forall t_m < t_i \quad (13)$$

Computing this sum empirically, we can check this property at $m = 50$ and $i = 100$ where we find

$$\frac{1}{2500} \sum_{l=1}^{2500} r^l(t_{100}) r^l(t_{50})^\top = \begin{bmatrix} 0.1955 & 0.0417 \\ -0.0504 & -0.1736 \end{bmatrix} \times 10^{-5} \quad (14)$$

All values in this matrix are close to zero, proving the independence of our residuals.

5 Conclusion

Overall, the Kalman filter that was developed was able to accurately determine the state and sensor bias from accelerometer and GPS measurements. For a given trial of the acceleration model, the error between the estimate and true state is minimal throughout a 30 second period. Moreover, by incorporating the accelerometer bias into our state vector, it was proven that our filter was able to determine the bias to a great degree of accuracy.

The empirical results from an ensemble of realizations also match with the theoretical properties of the Kalman filter, where we demonstrated that the conditional mean and error variance aligned with the projections of the Kalman filter. Moreover, checks on orthogonality and independence were performed and verified the results of our algorithm.